PROJECT DESIGN PHASE-II SPECIFICATION

StaffSync HRMS — Solution Requirements Specification

Complete functional (FR) and non-functional (NFR) engineering specifications for production deployment.

Project Name	**StaffSync — Enterprise Human Resource Management System**
Version & Status	v1.0.0 (Production Verified) | Lead: Parikshit Sharma
Live Endpoints	Client: staffsync-smoky.vercel.app | API: staffsync-czzq.onrender.com

1. Functional Requirements (FR)

FR No.	Epic / Category	Sub Requirement (Story / Sub-Task)	Implementation Evidence
FR-1	**User Authentication & RBAC**	• Secure Sign Up with department & role assignment • Password login with salted bcryptjs hashing • Signed JWT token generation (7-day duration) • Role guards for hr, manager, and employee	`authController.js` `POST /api/auth/login`
FR-2	**Workforce Directory**	• Search colleagues by name, email, or designation • Filter directory by Department & Status • Comprehensive employee dossier view • HR Admin CRUD controls to add/edit/delete personnel	`employeeController.js` `GET /api/employees`

FR-3	**Leave Request & Balances**	• Track allowances (Annual, Sick, Casual, Unpaid)
		• Pre-submission balance sufficiency check
		• Date range overlap collision prevention
		• Manager review desk with automatic balance deduction
			`leaveController.js`
			`POST /api/leaves`
			`PUT /api/leaves/:id/status`

FR-4	**Attendance Telemetry**	• 1-click arrival telemetry (Clock In) • Active session stopwatch counter • Shift completion (Clock Out) with auto-calculated hours • Daily presence registers (Present, Late, Half Day, Absent)	`attendanceController.js` `POST /api/attendance/clock-in`
FR-5	**Organizational Comms**	• Priority-classified bulletins (High, Medium, General) • Target audience filtering (All, hr, manager, employee) • High-priority visual alert banner on dashboards	`announcementController.js` `POST /api/announcements`
FR-6	**Learning & Upskilling**	• Curriculum tracks with duration & module counts • Full role parity: HR and Managers create tracks • Cohort enrollment & milestone progress updating	`trainingController.js` `POST /api/training`
FR-7	**Workforce Analytics**	• Aggregated organizational headcount by department • Attendance consistency and punctuality percentages • Leave utilization volumes and pending request ratios	`reportController.js` `GET /api/reports/dashboard`

2. Non-Functional Requirements (NFR)

NFR No.	Non-Functional Category	Description & Technical Benchmark	Production Compliance
NFR-1	**Usability & UX**	Editorial Warm Ivory (#EAE9DF) and Evergreen (#173D32) aesthetic with DM Serif Display headings and Inter body copy. Fully responsive layout.	Tailwind CSS v4 design tokens, Lucide icons, Three.js 3D canvas.
NFR-2	**Security & Cryptography**	All data in transit protected via TLS/HTTPS. Passwords hashed with bcryptjs (work factor 10). Stateless HMAC SHA-256 JWTs. CORS origin whitelist. Strict anti-mock policy.	bcryptjs, jsonwebtoken, CORS whitelist in server.js.
NFR-3	**Reliability & Integrity**	High data consistency via MongoDB Atlas replica sets. Atomic leave balance deductions prevent race conditions and over-deductions.	Mongoose schemas with indexed foreign keys & validation.
NFR-4	**Performance & Cold-Start**	Vercel CDN edge asset delivery under 80ms. Warm API response under 200ms. Client timeout calibrated to 60s for Render free-tier cold-start resilience.	Vite 8 minification, 60s timeout in apiClient.js.
NFR-5	**Availability & SLA**	Distributed architecture maintains 99.9% uptime SLA across Vercel Global Edge and Render Cloud Service with /api/health monitoring.	Vercel Edge Network + Render Container Service.
NFR-6	**Scalability & Modularity**	Decoupled 3-tier architecture separates UI from REST API business logic and persistence layer. Modular controllers and routes allow rapid scaling.	Express modular routers + React 19 component hierarchy.